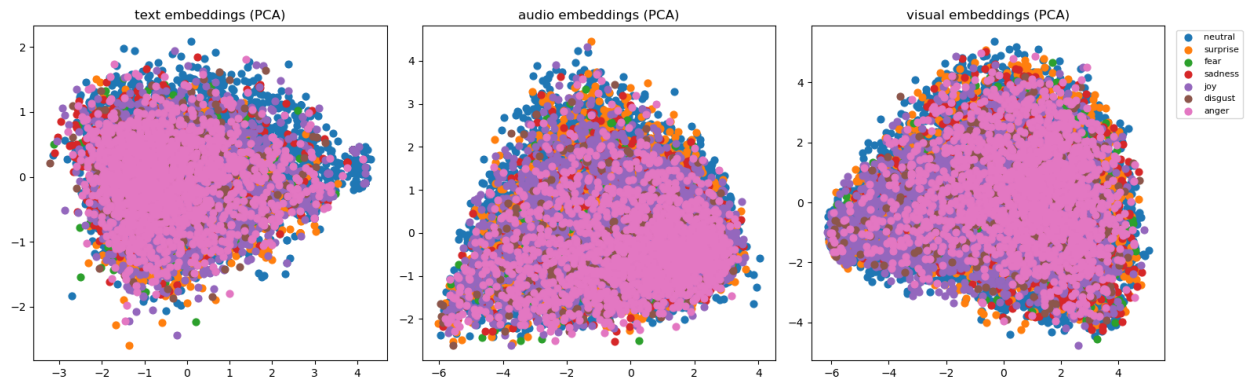


Data Audit

===== Split: train ===== csv: D:\University-Masters\Semester 3\P1 Affective Computing\Lab\Project\MELD\Dataset\MELD.Raw\train_sent_emo.csv video_dir: D:\University-Masters\Semester 3\P1 Affective Computing\Lab\Project\MELD\Dataset\MELD.Raw\train_splits rows in csv: 9989 OK: all 9989 rows have a matching .mp4 Emotion distribution: <table><tr><td>neutral</td><td>4710</td><td>(47.2%)</td></tr><tr><td>joy</td><td>1743</td><td>(17.4%)</td></tr><tr><td>surprise</td><td>1205</td><td>(12.1%)</td></tr><tr><td>anger</td><td>1109</td><td>(11.1%)</td></tr><tr><td>sadness</td><td>683</td><td>(6.8%)</td></tr><tr><td>disgust</td><td>271</td><td>(2.7%)</td></tr><tr><td>fear</td><td>268</td><td>(2.7%)</td></tr></table> Sentiment distribution: <table><tr><td>neutral</td><td>4710</td><td>(47.2%)</td></tr><tr><td>negative</td><td>2945</td><td>(29.5%)</td></tr><tr><td>positive</td><td>2334</td><td>(23.4%)</td></tr></table> Utterance length (words): mean=8.0, max=69, 95th pct=20 9989 .mp4 files on disk, 7.88 GB total	neutral	4710	(47.2%)	joy	1743	(17.4%)	surprise	1205	(12.1%)	anger	1109	(11.1%)	sadness	683	(6.8%)	disgust	271	(2.7%)	fear	268	(2.7%)	neutral	4710	(47.2%)	negative	2945	(29.5%)	positive	2334	(23.4%)	===== Split: dev ===== csv: D:\University-Masters\Semester 3\P1 Affective Computing\Lab\Project\MELD\Dataset\MELD.Raw\dev_sent_emo.csv video_dir: D:\University-Masters\Semester 3\P1 Affective Computing\Lab\Project\MELD\Dataset\MELD.Raw\dev_splits rows in csv: 1109 !! 1 / 1109 rows have NO matching .mp4 first few missing: ['dia110_utt7.mp4'] Emotion distribution: <table><tr><td>neutral</td><td>470</td><td>(42.4%)</td></tr><tr><td>joy</td><td>163</td><td>(14.7%)</td></tr><tr><td>anger</td><td>153</td><td>(13.8%)</td></tr><tr><td>surprise</td><td>150</td><td>(13.5%)</td></tr><tr><td>sadness</td><td>111</td><td>(10.0%)</td></tr><tr><td>fear</td><td>40</td><td>(3.6%)</td></tr><tr><td>disgust</td><td>22</td><td>(2.0%)</td></tr></table> Sentiment distribution: <table><tr><td>neutral</td><td>470</td><td>(42.4%)</td></tr><tr><td>negative</td><td>406</td><td>(36.6%)</td></tr><tr><td>positive</td><td>233</td><td>(21.0%)</td></tr></table> Utterance length (words): mean=7.9, max=37, 95th pct=20 1112 .mp4 files on disk, 0.78 GB total	neutral	470	(42.4%)	joy	163	(14.7%)	anger	153	(13.8%)	surprise	150	(13.5%)	sadness	111	(10.0%)	fear	40	(3.6%)	disgust	22	(2.0%)	neutral	470	(42.4%)	negative	406	(36.6%)	positive	233	(21.0%)	===== Split: test ===== csv: D:\University-Masters\Semester 3\P1 Affective Computing\Lab\Project\MELD\Dataset\MELD.Raw\test_sent_emo.csv video_dir: D:\University-Masters\Semester 3\P1 Affective Computing\Lab\Project\MELD\Dataset\MELD.Raw\test_splits rows in csv: 2610 OK: all 2610 rows have a matching .mp4 Emotion distribution: <table><tr><td>neutral</td><td>1256</td><td>(48.1%)</td></tr><tr><td>joy</td><td>402</td><td>(15.4%)</td></tr><tr><td>anger</td><td>345</td><td>(13.2%)</td></tr><tr><td>surprise</td><td>281</td><td>(10.8%)</td></tr><tr><td>sadness</td><td>208</td><td>(8.0%)</td></tr><tr><td>disgust</td><td>68</td><td>(2.6%)</td></tr><tr><td>fear</td><td>50</td><td>(1.9%)</td></tr></table> Sentiment distribution: <table><tr><td>neutral</td><td>1256</td><td>(48.1%)</td></tr><tr><td>negative</td><td>833</td><td>(31.9%)</td></tr><tr><td>positive</td><td>521</td><td>(20.0%)</td></tr></table> Utterance length (words): mean=8.2, max=45, 95th pct=21 4807 .mp4 files on disk, 2.24 GB total	neutral	1256	(48.1%)	joy	402	(15.4%)	anger	345	(13.2%)	surprise	281	(10.8%)	sadness	208	(8.0%)	disgust	68	(2.6%)	fear	50	(1.9%)	neutral	1256	(48.1%)	negative	833	(31.9%)	positive	521	(20.0%)
neutral	4710	(47.2%)																																																																																										
joy	1743	(17.4%)																																																																																										
surprise	1205	(12.1%)																																																																																										
anger	1109	(11.1%)																																																																																										
sadness	683	(6.8%)																																																																																										
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joy	163	(14.7%)																																																																																										
anger	153	(13.8%)																																																																																										
surprise	150	(13.5%)																																																																																										
sadness	111	(10.0%)																																																																																										
fear	40	(3.6%)																																																																																										
disgust	22	(2.0%)																																																																																										
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negative	406	(36.6%)																																																																																										
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negative	833	(31.9%)																																																																																										
positive	521	(20.0%)																																																																																										

Project Outputs

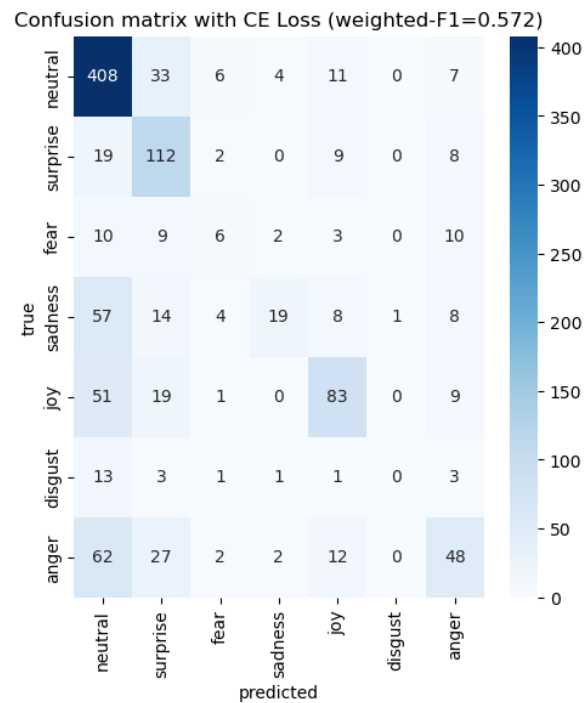
|-- 2D_embeddings.png



-- attention_fusion_ce_accuracy_loss.png



-- attention_fusion_ce_cm.png



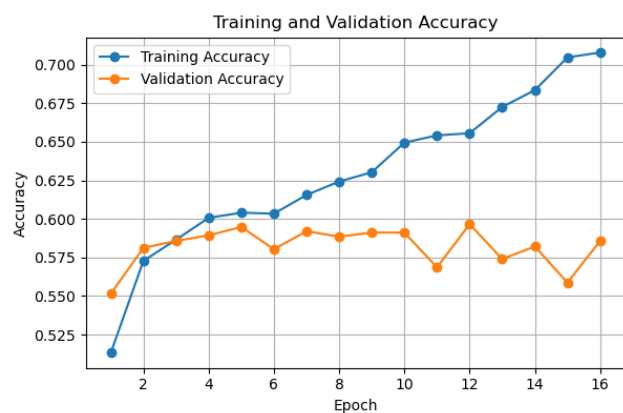
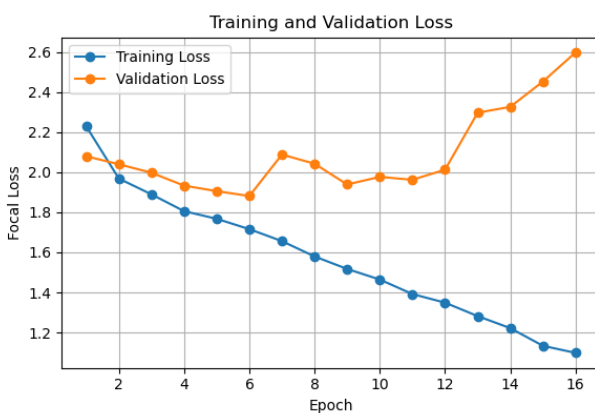
Dev classification_report with Cross-Entropy Loss: 0.5723

precision recall f1-score support

neutral	0.6581	0.8699	0.7493	469
surprise	0.5161	0.7467	0.6104	150
fear	0.2727	0.1500	0.1935	40
sadness	0.6786	0.1712	0.2734	111
joy	0.6535	0.5092	0.5724	163
disgust	0.0000	0.0000	0.0000	22
anger	0.5161	0.3137	0.3902	153

accuracy		0.6101	1108	
macro avg	0.4707	0.3944	0.3985	1108
weighted avg	0.5937	0.6101	0.5723	1108

|-- attention_fusion_focal_accuracy_loss.png



|-- attention_fusion_focal_cm.png

Dev classification_report with Focal Loss: 0.5779

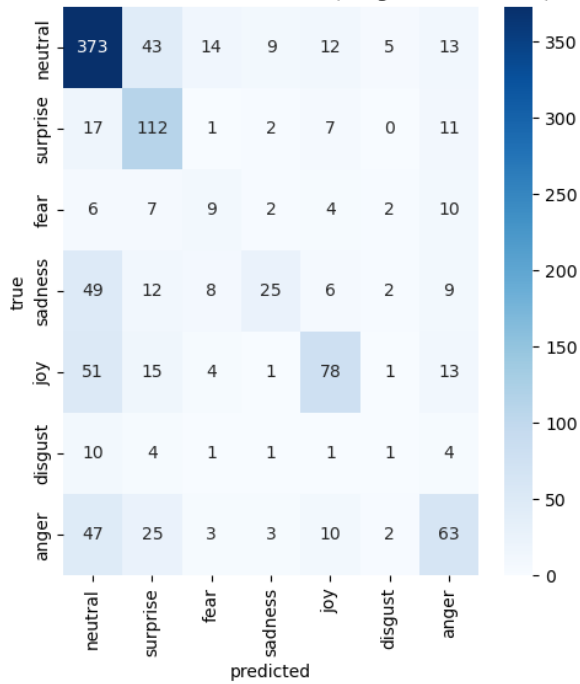
precision recall f1-score support

neutral	0.6745	0.7953	0.7299	469
surprise	0.5138	0.7467	0.6087	150
fear	0.2250	0.2250	0.2250	40
sadness	0.5814	0.2252	0.3247	111
joy	0.6610	0.4785	0.5552	163
disgust	0.0769	0.0455	0.0571	22
anger	0.5122	0.4118	0.4565	153

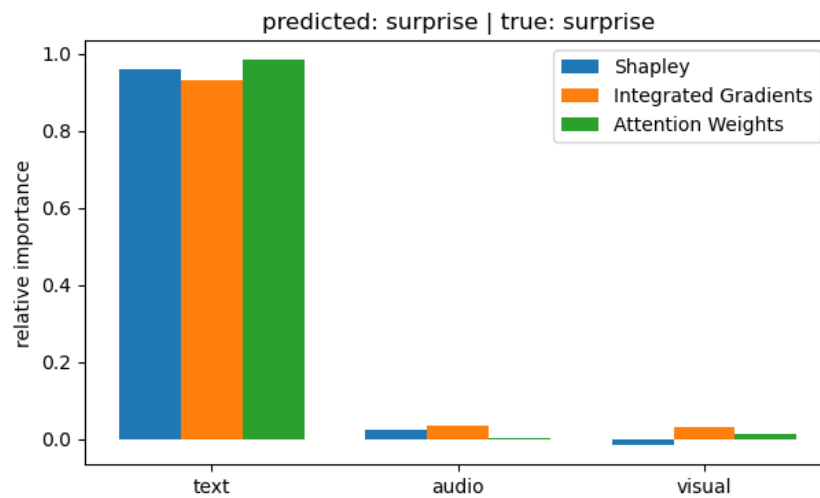
accuracy		0.5966	1108	
macro avg	0.4635	0.4183	0.4224	1108

weighted avg	0.5909	0.5966	0.5779	1108
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Confusion matrix with Focal Loss (weighted-F1=0.578)



|-- attention_fusion_modality_agreement_cross_entropy.png



{	'text': 0.9584799179953449,	
	'audio': 0.02464270893263731,	
	'visual': -0.016877373072017747},	
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{'text': 0.983688473701477,  
'audio': 0.002630524104461074,  
'visual': 0.013681101612746716}}
```

Shapley vs IG agreement: 19/20 (95%)

Shapley vs Classifier agreement: 19/20 (95%)

IG vs Classifier agreement: 20/20 (100%)

All three agree: 19/20 (95%)

```
[OK] utt 0: predicted=surprise shapley_top=text ig_top=text fusion_top=text  
[OK] utt 1: predicted=surprise shapley_top=text ig_top=text fusion_top=text  
[OK] utt 2: predicted=neutral shapley_top=text ig_top=text fusion_top=text  
[OK] utt 3: predicted=fear shapley_top=text ig_top=text fusion_top=text  
[OK] utt 4: predicted=neutral shapley_top=text ig_top=text fusion_top=text  
[OK] utt 5: predicted=surprise shapley_top=text ig_top=text fusion_top=text  
[OK] utt 6: predicted=neutral shapley_top=text ig_top=text fusion_top=text  
[OK] utt 7: predicted=neutral shapley_top=text ig_top=text fusion_top=text  
[OK] utt 8: predicted=neutral shapley_top=text ig_top=text fusion_top=text  
[OK] utt 9: predicted=neutral shapley_top=text ig_top=text fusion_top=text  
[OK] utt 10: predicted=neutral shapley_top=text ig_top=text fusion_top=text  
[OK] utt 11: predicted=joy shapley_top=text ig_top=text fusion_top=text  
[OK] utt 12: predicted=neutral shapley_top=text ig_top=text fusion_top=text  
[OK] utt 13: predicted=surprise shapley_top=text ig_top=text fusion_top=text  
[OK] utt 14: predicted=anger shapley_top=text ig_top=text fusion_top=text  
[OK] utt 15: predicted=joy shapley_top=text ig_top=text fusion_top=text  
[--] utt 16: predicted=surprise shapley_top=visual ig_top=text fusion_top=text  
[OK] utt 17: predicted=joy shapley_top=text ig_top=text fusion_top=text  
[OK] utt 18: predicted=anger shapley_top=text ig_top=text fusion_top=text  
[OK] utt 19: predicted=anger shapley_top=text ig_top=text fusion_top=text
```

|-- attention_fusion_modality_agreement_focal.png

```
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{'text': 0.7150276407197454,  
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'visual': 0.13984145861109776},  
{'text': 0.9023736715316772,  
'audio': 0.010926393792033195,  
'visual': 0.08670002222061157}})
```

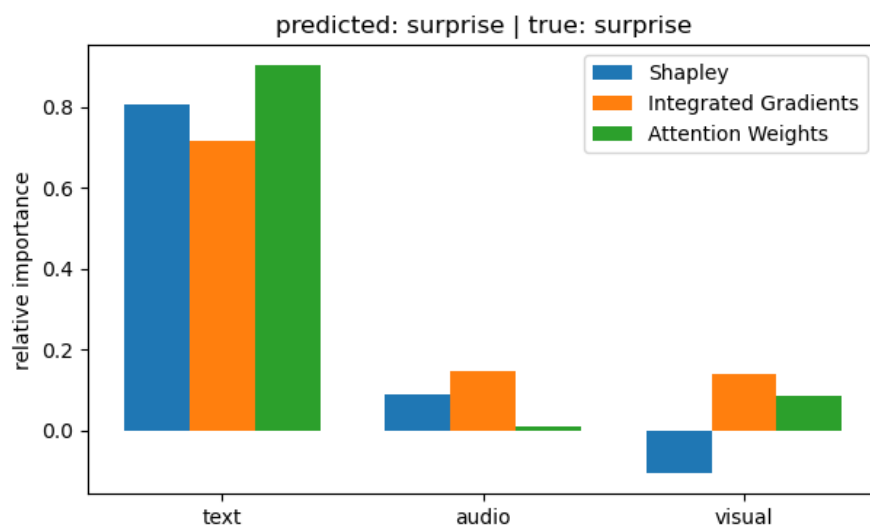
Shapley vs IG agreement: 15/20 (75%)

Shapley vs Classifier agreement: 19/20 (95%)

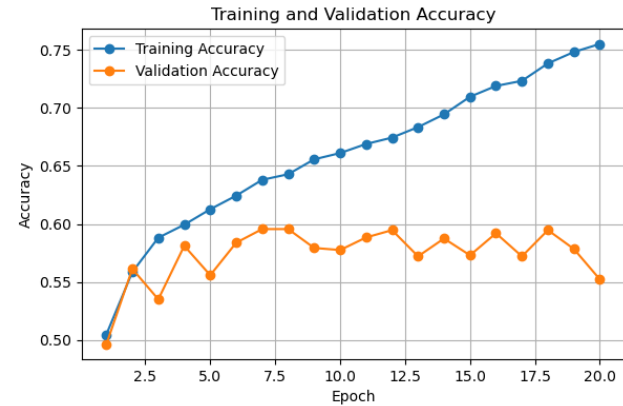
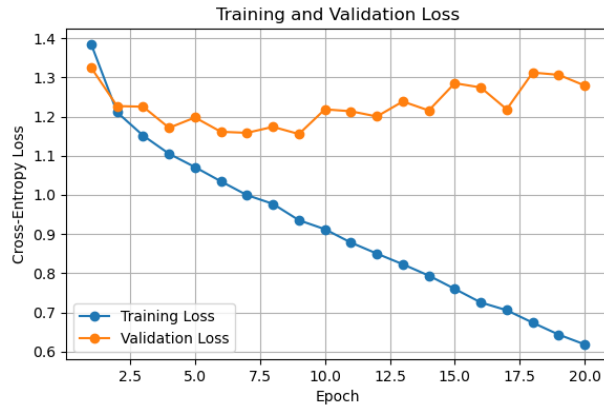
IG vs Classifier agreement: 14/20 (70%)

All three agree: 14/20 (70%)

[OK]	utt 0:	predicted=surprise	shapley_top=text	ig_top=text	fusion_top=text
[OK]	utt 1:	predicted=surprise	shapley_top=text	ig_top=text	fusion_top=text
[OK]	utt 2:	predicted=neutral	shapley_top=text	ig_top=text	fusion_top=text
[--]	utt 3:	predicted=sadness	shapley_top=text	ig_top=visual	fusion_top=text
[OK]	utt 4:	predicted=neutral	shapley_top=text	ig_top=text	fusion_top=text
[OK]	utt 5:	predicted=surprise	shapley_top=text	ig_top=text	fusion_top=text
[OK]	utt 6:	predicted=neutral	shapley_top=text	ig_top=text	fusion_top=text
[OK]	utt 7:	predicted=neutral	shapley_top=text	ig_top=text	fusion_top=text
[OK]	utt 8:	predicted=neutral	shapley_top=text	ig_top=text	fusion_top=text
[OK]	utt 9:	predicted=neutral	shapley_top=text	ig_top=text	fusion_top=text
[OK]	utt 10:	predicted=neutral	shapley_top=text	ig_top=text	fusion_top=text
[--]	utt 11:	predicted=joy	shapley_top=text	ig_top=visual	fusion_top=text
[OK]	utt 12:	predicted=neutral	shapley_top=text	ig_top=text	fusion_top=text
[OK]	utt 13:	predicted=surprise	shapley_top=text	ig_top=text	fusion_top=text
[OK]	utt 14:	predicted=fear	shapley_top=text	ig_top=text	fusion_top=text
[--]	utt 15:	predicted=joy	shapley_top=text	ig_top=visual	fusion_top=text
[--]	utt 16:	predicted=joy	shapley_top=visual	ig_top=visual	fusion_top=text
[--]	utt 17:	predicted=joy	shapley_top=text	ig_top=visual	fusion_top=text
[--]	utt 18:	predicted=anger	shapley_top=text	ig_top=visual	fusion_top=text
[OK]	utt 19:	predicted=fear	shapley_top=text	ig_top=text	fusion_top=text



|-- linear_fusion_ce_accuracy_loss.png



```

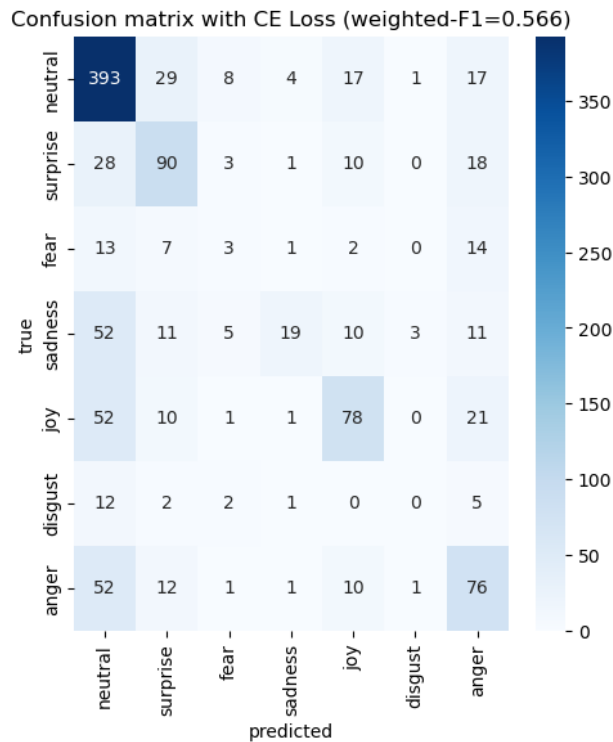
epoch 1 | train_loss=1.3852 | val_loss=1.3250 | val_weighted_f1=0.3928 |
val_macro_f1=0.2029
epoch 2 | train_loss=1.2109 | val_loss=1.2267 | val_weighted_f1=0.4878 |
val_macro_f1=0.2929
epoch 3 | train_loss=1.1515 | val_loss=1.2251 | val_weighted_f1=0.4451 |
val_macro_f1=0.2601
epoch 4 | train_loss=1.1050 | val_loss=1.1712 | val_weighted_f1=0.5284 |
val_macro_f1=0.3331
epoch 5 | train_loss=1.0709 | val_loss=1.1976 | val_weighted_f1=0.5033 |
val_macro_f1=0.3295
epoch 6 | train_loss=1.0345 | val_loss=1.1611 | val_weighted_f1=0.5423 |
val_macro_f1=0.3595
epoch 7 | train_loss=0.9995 | val_loss=1.1584 | val_weighted_f1=0.5485 |
val_macro_f1=0.3591
epoch 8 | train_loss=0.9770 | val_loss=1.1737 | val_weighted_f1=0.5518 |
val_macro_f1=0.3712
epoch 9 | train_loss=0.9353 | val_loss=1.1554 | val_weighted_f1=0.5447 |
val_macro_f1=0.3655
epoch 10 | train_loss=0.9125 | val_loss=1.2184 | val_weighted_f1=0.5269 |
val_macro_f1=0.3500
epoch 11 | train_loss=0.8784 | val_loss=1.2135 | val_weighted_f1=0.5372 |
val_macro_f1=0.3467
epoch 12 | train_loss=0.8502 | val_loss=1.2001 | val_weighted_f1=0.5585 |
val_macro_f1=0.3756
epoch 13 | train_loss=0.8229 | val_loss=1.2384 | val_weighted_f1=0.5413 |
val_macro_f1=0.3804
epoch 14 | train_loss=0.7940 | val_loss=1.2155 | val_weighted_f1=0.5600 |
val_macro_f1=0.3846
epoch 15 | train_loss=0.7601 | val_loss=1.2851 | val_weighted_f1=0.5419 |
val_macro_f1=0.3629
epoch 16 | train_loss=0.7255 | val_loss=1.2742 | val_weighted_f1=0.5599 |
val_macro_f1=0.3796

```

```
epoch 17 | train_loss=0.7057 | val_loss=1.2184 | val_weighted_f1=0.5533 |
val_macro_f1=0.3911
epoch 18 | train_loss=0.6739 | val_loss=1.3122 | val_weighted_f1=0.5656 |
val_macro_f1=0.3860
epoch 19 | train_loss=0.6435 | val_loss=1.3065 | val_weighted_f1=0.5391 |
val_macro_f1=0.3656
epoch 20 | train_loss=0.6184 | val_loss=1.2792 | val_weighted_f1=0.5300 |
val_macro_f1=0.3744

best val weighted-F1: 0.5656
```

|-- linear_fusion_ce_cm.png



Dev classification_report with Cross-Entropy Loss: 0.5656

	precision	recall	f1-score	support
--	-----------	--------	----------	---------

neutral	0.6528	0.8380	0.7339	469
surprise	0.5590	0.6000	0.5788	150
fear	0.1304	0.0750	0.0952	40
sadness	0.6786	0.1712	0.2734	111
joy	0.6142	0.4785	0.5379	163
disgust	0.0000	0.0000	0.0000	22
anger	0.4691	0.4967	0.4825	153

accuracy		0.5948	1108
macro avg	0.4434	0.3799	0.3860
weighted avg	0.5798	0.5948	0.5656

|-- linear_fusion_focal_accuracy_loss.png

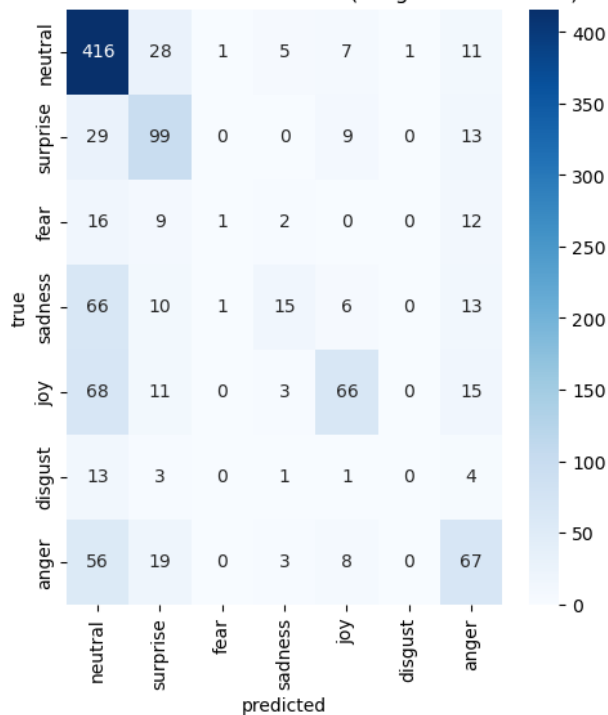


epoch 1 | train_loss=2.2699 | val_loss=2.2496 | val_weighted_f1=0.4084 | val_macro_f1=0.2192
epoch 2 | train_loss=1.9572 | val_loss=2.0300 | val_weighted_f1=0.4813 | val_macro_f1=0.2882
epoch 3 | train_loss=1.8428 | val_loss=2.0390 | val_weighted_f1=0.4380 | val_macro_f1=0.2575
epoch 4 | train_loss=1.7505 | val_loss=1.9570 | val_weighted_f1=0.5283 | val_macro_f1=0.3339
epoch 5 | train_loss=1.6818 | val_loss=2.0126 | val_weighted_f1=0.4839 | val_macro_f1=0.3079
epoch 6 | train_loss=1.6091 | val_loss=1.9151 | val_weighted_f1=0.5401 | val_macro_f1=0.3740
epoch 7 | train_loss=1.5464 | val_loss=1.9248 | val_weighted_f1=0.5544 | val_macro_f1=0.3671
epoch 8 | train_loss=1.4964 | val_loss=1.9780 | val_weighted_f1=0.5423 | val_macro_f1=0.3762
epoch 9 | train_loss=1.4311 | val_loss=1.9264 | val_weighted_f1=0.5488 | val_macro_f1=0.3762
epoch 10 | train_loss=1.3658 | val_loss=2.0462 | val_weighted_f1=0.5244 | val_macro_f1=0.3472
epoch 11 | train_loss=1.3130 | val_loss=2.0718 | val_weighted_f1=0.5431 | val_macro_f1=0.3467
no improvement in 4 epochs, stopping early

best val weighted-F1: 0.5544

|-- linear_fusion_focal_cm.png

Confusion matrix with Focal Loss (weighted-F1=0.554)



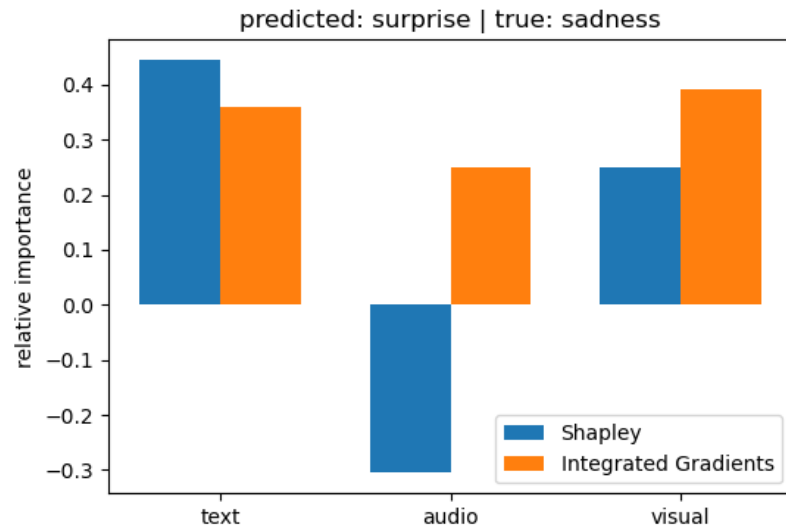
Dev classification_report with Focal Loss: 0.5544

	precision	recall	f1-score	support
--	-----------	--------	----------	---------

neutral	0.6265	0.8870	0.7343	469
surprise	0.5531	0.6600	0.6018	150
fear	0.3333	0.0250	0.0465	40
sadness	0.5172	0.1351	0.2143	111
joy	0.6804	0.4049	0.5077	163
disgust	0.0000	0.0000	0.0000	22
anger	0.4963	0.4379	0.4653	153

accuracy		0.5993		1108
macro avg	0.4581	0.3643	0.3671	1108
weighted avg	0.5725	0.5993	0.5544	1108

|-- linear_fusion_modality_agreement_cross_entropy.png

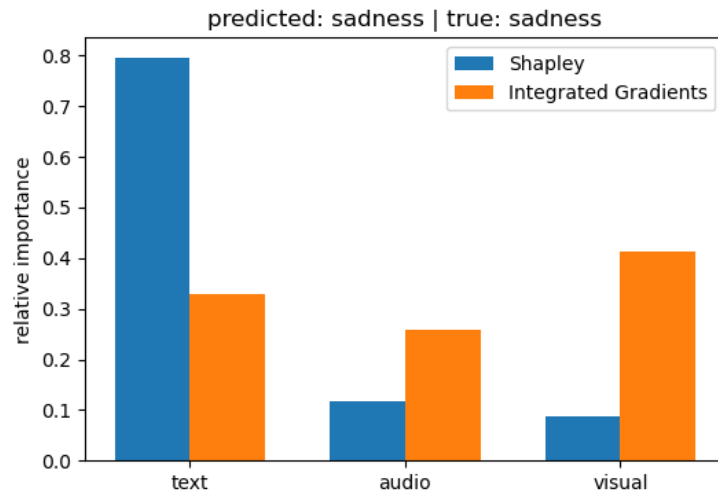


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{'text': 0.4442195813021344,
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 'visual': 0.2506396069026879},
{'text': 0.35838333381025683,
 'audio': 0.2501608941829194,
 'visual': 0.3914557720068238})
```

top-modality agreement: 7/10 utterances (70%)

```
[--] utt 0: predicted=surprise shapley_top=text ig_top=visual
[OK] utt 1: predicted=neutral shapley_top=visual ig_top=visual
[OK] utt 2: predicted=anger shapley_top=visual ig_top=visual
[OK] utt 3: predicted=neutral shapley_top=visual ig_top=visual
[--] utt 4: predicted=surprise shapley_top=text ig_top=visual
[OK] utt 5: predicted=neutral shapley_top=visual ig_top=visual
[--] utt 6: predicted=joy shapley_top=text ig_top=visual
[OK] utt 7: predicted=neutral shapley_top=visual ig_top=visual
[OK] utt 8: predicted=neutral shapley_top=visual ig_top=visual
[OK] utt 9: predicted=neutral shapley_top=visual ig_top=visual
```

|-- linear_fusion_modality_agreement_focal.png

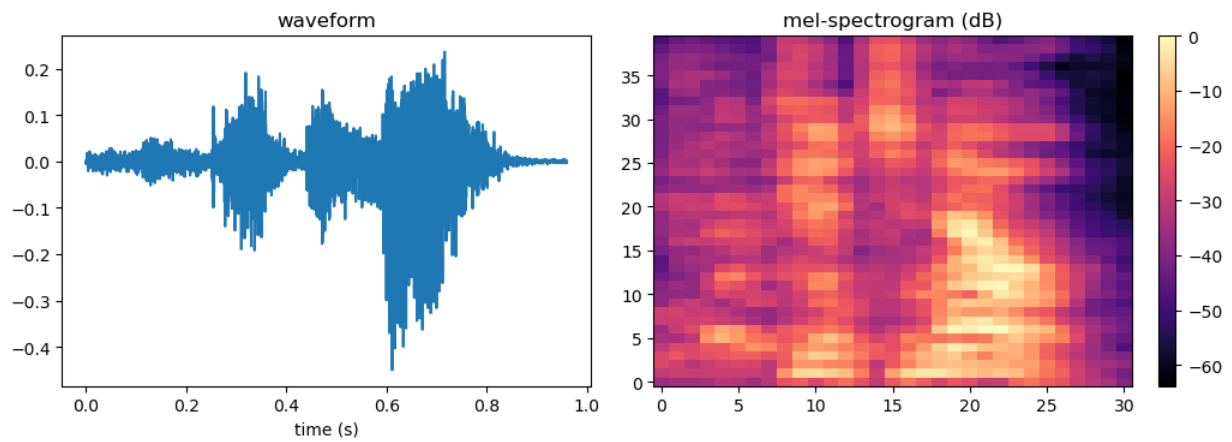


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{'text': 0.7952657035919319,
 'audio': 0.11771173714840315,
 'visual': 0.08702255925966483},
{'text': 0.3286287908383362,
 'audio': 0.25834374875594573,
 'visual': 0.4130274604057181})
```

top-modality agreement: 7/10 utterances (70%)

```
[OK] utt 0: predicted=neutral  shapley_top=text  ig_top=text
[OK] utt 1: predicted=anger    shapley_top=visual ig_top=visual
[--] utt 2: predicted=neutral  shapley_top=audio  ig_top=text
[OK] utt 3: predicted=anger    shapley_top=visual ig_top=visual
[--] utt 4: predicted=neutral  shapley_top=audio  ig_top=text
[OK] utt 5: predicted=joy      shapley_top=text  ig_top=text
[--] utt 6: predicted=neutral  shapley_top=visual ig_top=text
[OK] utt 7: predicted=neutral  shapley_top=visual ig_top=visual
[OK] utt 8: predicted=surprise shapley_top=text  ig_top=text
[OK] utt 9: predicted=surprise shapley_top=text  ig_top=text
```

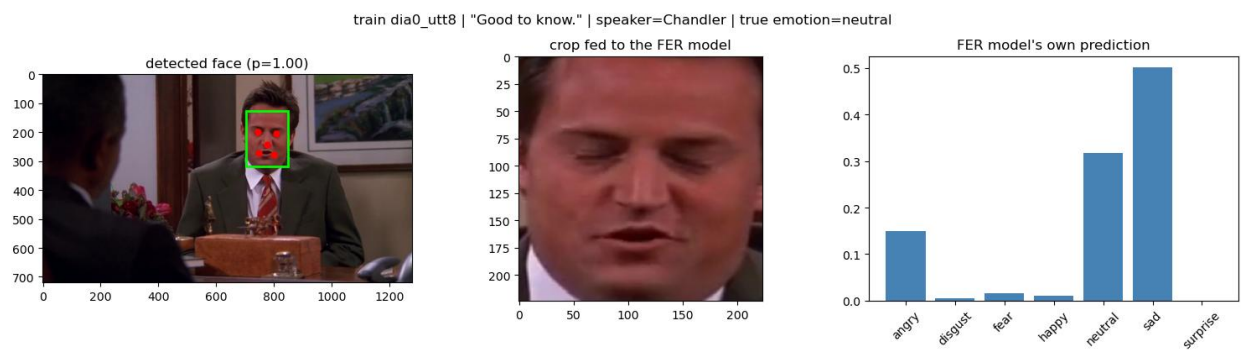
|-- visual_audio_embeddings.png



|-- visual_text_embeddings.png



|__ visual_video_embeddings.png



Explanation Report

Case 1: Less Accuracy, Better Explainability

(Linear Classifier (without dropout and without returning the weights), Focal Loss (for LLM explanation), Batch = 8, Epoch=20, alpha=ALPHA_SQRT, per-sample L2-normalized embeddings, Patience = 4, LR = 1e-3, Gamma = 1)

Shapely	Integrated Gradient
--- dia0_utt0 --- utterance: "Oh my God, he's lost it. He's totally lost it." predicted: sadness true: sadness The model likely made this prediction because Phoebe's statement "he's lost it" is a phrase that can be interpreted as both a expression of frustration and disappointment, but also a sense of sadness. The fact that Phoebe uses the phrase in a way that suggests a strong emotional reaction, with the intensity implied by the phrase, leads the model to infer that she is primarily feeling sadness, rather than just frustration. The model is picking up on the tone and emotional intensity of Phoebe's words, which is heavily weighted towards the text modality, to make this prediction. --- dia0_utt1 --- utterance: "What?" predicted: surprise true: surprise When Monica says "What?", the model likely made a surprise prediction because her tone of voice, which is conveyed through the audio modality, is quite abrupt and inquiring. This tone is more typical of surprise or questioning than a neutral tone. The model's analysis suggests that this audio contribution was the primary driver of the surprise prediction, outweighing the more neutral language in her words ("What?"). The slight negative impact of the text and visual components may have been too small to change the overall direction of the prediction. --- dia100_utt0 --- utterance: "Okay. It's Emma." predicted: neutral true: neutral	--- dia0_utt0 --- utterance: "Oh my God, he's lost it. He's totally lost it." predicted: sadness true: sadness Phoebe's strong emotional language, particularly the phrase "he's totally lost it," suggests that the model is picking up on a sense of desperation or urgency. The text modality, which includes the words she chooses, is driving the model's prediction of sadness, as it conveys a sense of distress and concern. The intensity of the language, rather than the tone or facial expression, seems to be the key factor in the model's decision. --- dia0_utt1 --- utterance: "What?" predicted: surprise true: surprise When Monica said "What?", the model likely predicted surprise because the text modality, specifically the words she chose, played a significant role in this decision. The phrase "What?" is often used as an expression of surprise or inquiry, and its presence in the input likely triggered the surprise emotion. In contrast, the audio and visual modalities, although present, didn't seem to contribute as strongly to the model's decision, suggesting that the emotional cue was mainly derived from the text itself. --- dia100_utt0 --- utterance: "Okay. It's Emma." predicted: neutral true: neutral When the model predicted Emma's emotion as neutral, it likely did so because the text modality (the words "It's Emma") had a relatively low impact on the prediction. The audio modality, which includes the tone and pitch of Monica's voice, might have

When the model predicts that the emotion is neutral, it's likely because the text input (what Monica said) only contains a single name, "Emma", without any additional context or sentiment. This lack of emotional cue suggests that the model relied heavily on the text modality to make the prediction, likely due to its 28% contribution to the decision. --- dia101_utt0 --- utterance: "Emma! See? I don't want it." predicted: anger true: sadness The model likely predicted anger because Rachel's facial expression, not her verbal statement, is the dominant contributor to the prediction. The fact that the audio modality is negatively valued (-26%) suggests that the model considered the tone or volume of Rachel's voice to be somewhat critical or forceful, which is consistent with an angry emotional state. This contrast with the higher value assigned to the text modality (18%), which may indicate that the model thought Rachel's words were more neutral or non-confrontational, but the overall prediction is driven by the facial expression, which conveyed a stronger emotional signal. --- dia102_utt0 --- utterance: "Take it." predicted: neutral true: neutral The model likely predicted a neutral emotion for Monica's utterance because the text modality played a significant role in this prediction. The model picked up on the straightforward and practical tone of the words "Take it," which didn't convey any strong emotions like excitement, surprise, or frustration. This suggests that the text modality, particularly the words and their meaning, dominated the prediction, making the emotion appear neutral.	suggested a slightly more neutral or even surprised tone, but it wasn't enough to override the neutral feeling conveyed by the text. The visual modality, which isn't directly provided in this context, might have added to the neutral feeling by implying a straightforward statement rather than a more emotional or expressive one. --- dia101_utt0 --- utterance: "Emma! See? I don't want it." predicted: anger true: sadness Based on the predicted emotion of anger, it's likely that the model's decision was heavily influenced by the audio modality, as the speaker's tone and pace suggest a sense of frustration or annoyance. The model may have picked up on the speaker's raised pitch and slightly emphatic words, which typically accompany angry outbursts. The text modality, while also contributing to the prediction, likely played a supporting role, as the words themselves don't necessarily convey a strong emotion. The visual modality is less likely to have driven the decision, as there's no explicit information about Emma's facial expression or body language that would have led the model to predict anger. --- dia102_utt0 --- utterance: "Take it." predicted: neutral true: neutral The model likely made this prediction because Monica's neutral tone and the simple, straightforward phrase "Take it" don't convey a strong emotional message. The fact that the text modality contributes a significant 38% to the prediction suggests that the model gave more weight to the words themselves, as opposed to the tone or facial expression that might have accompanied them. This is because the model is focusing on the literal meaning of
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	the words, which in this case doesn't suggest a strong emotional response.
Case 2: More Accuracy, Biased Explainability (Attention-based Classifier (dropout + soft floor + no entropy penalty), Focal Loss (for LLM explanation), Batch = 8, Epoch=20, alpha=ALPHA_SQRT, global per-modality rescaling, Patience = 4, LR = 1e-3, Gamma = 1)	
Shapely	Integrated Gradient
--- dia0_utt0 --- utterance: "Oh my God, he's lost it. He's totally lost it." predicted: surprise true: sadness Phoebe's statement suggests that someone is completely overwhelmed or shocked, and the model likely predicted surprise because of the intensity and certainty of her words. The fact that Phoebe uses the phrase "totally lost it," which implies a strong emotional state, is likely what drove the decision. The text modality, which carries the actual words, is likely the primary contributor to this prediction, as it conveys the emotional tone and intensity of Phoebe's statement. --- dia0_utt1 --- utterance: "What?" predicted: surprise true: surprise When Monica says "What?", the model likely made a surprise prediction because the text contribution is high, indicating that the words themselves play a significant role in the model's decision. Specifically, the text contribution is 81%, suggesting that the words "What?" are crucial in conveying a sense of questioning or inquiry, which is often accompanied by surprise. The model likely picked up on the uncertainty and unexpectedness of Monica's question, leading to the surprise emotion prediction. --- dia100_utt0 --- utterance: "Okay. It's Emma." predicted: neutral true: neutral So, when the model predicted "neutral" for Emma, it's likely because the text	--- dia0_utt0 --- utterance: "Oh my God, he's lost it. He's totally lost it." predicted: surprise true: sadness When Phoebe says "Oh my God, he's lost it," the model likely predicted surprise because the text modality plays a significant role in this prediction. The intensity of the language, with the use of "lost it" and "Oh my God," suggests a strong emotional response, which is more characteristic of surprise than other emotions. The model is likely drawing a connection between the intensity of Phoebe's words and the intensity of the speaker's emotion, leading to the surprise prediction. --- dia0_utt1 --- utterance: "What?" predicted: surprise true: surprise When Monica says "What?", the model is surprised because the context of the conversation was likely established before this question. This suggests that the model is interpreting Monica's question as a deviation from the expected flow of conversation, which indicates a surprise or unexpected turn. The fact that the model was surprised by this utterance, rather than a neutral or expected one, suggests that the modality driving this decision was likely the text - the actual words Monica spoke. The text conveyed a sense of inquiry or inquiry in a new context, which the model likely picked up on to trigger the surprise emotion. --- dia100_utt0 ---

contribution to the prediction was significantly stronger than the other modality contributions. The text, which was the utterance "Okay", provided a clear and neutral statement that conveyed no strong emotion or sentiment. This strong text contribution likely overpowered the weaker visual and audio contributions, which were barely relevant to the prediction. --- dia101_utt0 --- utterance: "Emma! See? I don't want it." predicted: sadness true: sadness Based on the model's prediction, it's likely that the sadness was driven by the speaker's tone of voice, which is indicated by the "audio" contribution of -4%. Although Rachel's words "I don't want it" are positive, the negative emotional tone of her voice suggests a sense of resignation or disappointment, which is more closely aligned with sadness. This subtle cue from the audio modality likely played a significant role in the model's prediction, outweighing the more positive but less emotive words, and the positive emotional tone of the text and visual modality. --- dia102_utt0 --- utterance: "Take it." predicted: neutral true: neutral The model likely predicted Monica's emotion as neutral because the text, which is the spoken words, played a dominant role in this prediction. The model was heavily influenced by the words "Take it", which, taken literally, suggest a neutral or even slightly negative tone. The audio and visual modalities contributed less to the prediction, with the visual input being particularly negative (-22%), which may have indicated a slightly surprised or hesitant tone in Monica's voice, but it wasn't enough to tip the balance towards a different emotion.	utterance: "Okay. It's Emma." predicted: neutral true: neutral When the model predicted that Monica's emotion is neutral, it's likely because the text contributed the most to the prediction. The model took Monica's words, "It's Emma," at face value, which seemed like a straightforward statement about someone's identity. The text contribution was strong because it was a factual statement that didn't convey any strong emotions. The other modality contributions, including visual and audio, were relatively minor, suggesting that Monica's facial expression and tone of voice didn't strongly suggest a particular emotion. --- dia101_utt0 --- utterance: "Emma! See? I don't want it." predicted: sadness true: sadness Based on the model's prediction, it likely made the connection to sadness because Rachel's words, which were the dominant modality contributing to the prediction (41% of the total), expressed a lack of desire or willingness, which is a common emotional cue associated with sadness. The fact that the words were spoken in a tone that didn't convey strong emotion, but still conveyed a clear message of hesitation, likely reinforced the model's interpretation of sadness. The relatively minor contribution of the audio modality (10%) and the visual modality (50%) suggests that the spoken words were the most critical factor in the model's prediction, rather than the visual cues or the tone of voice itself. --- dia102_utt0 --- utterance: "Take it." predicted: neutral true: neutral The model predicted that Monica's emotion was neutral because the majority of the analysis (66%) was driven by the text component, which is likely the spoken words
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	"Take it." This suggests that the model focused more on the literal meaning of Monica's words, interpreting "Take it" as a neutral statement, rather than looking for emotional nuances in her tone or facial expression. The smaller contributions from visual (26%) and audio (8%) modalities likely didn't provide enough evidence to shift the model's prediction from neutral.
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